

Automated fMRI voxel selection for cerebrospinal fluid (CSF) inflow signal in the fourth ventricle

Category: General

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Introduction

fMRI acquired with high temporal resolution (<400 ms) can capture inflow of CSF in select voxels at the bottom of the fourth ventricle.¹ However, there is no automated way to select these inflow voxels. Existing brain segmentation tools are insufficient as they operate in anatomical space, whereas inflow voxels must be identified directly in the lower-resolution functional space. Therefore, researchers must manually inspect each fourth ventricle voxel, which is time-consuming and labor-intensive. Here, we build a system that automatically selects voxels in the fourth ventricle that contain CSF inflow signal, lowering the barrier for researchers to study CSF dynamics.

Data

Our model is trained on manually-defined masks from 292 fMRI scans across 77 participants (35 with age $\in [18, 39]$, 42 with age $\in [60, 83]$). These manual masks were obtained by visually inspecting the single band reference (SBRef) image acquired prior to each functional scan (i.e., an image acquired with the same functional parameters but no acceleration), as well as the BOLD timeseries of each voxel. We labeled voxels only in the bottom 4 slices, where the inflow signal is strongest.¹

All data were collected on a 3T MRI scanner. Anatomical images were acquired using 1mm isotropic T1-weighted multi-echo MPRAGE. Functional data were acquired using gradient-echo EPI (TR=0.378 s, TE=31 ms, flip angle=37°, 2.5 mm isotropic voxels, 40 slices, multiband factor = 8, blipped CAIPI shift=FOV/4, no in-plane acceleration), positioned such that the bottom slice intersects the fourth ventricle.¹

Methods

First, we apply the FreeSurfer automatic segmentation on the anatomical image and extract the fourth ventricle. This anatomical mask is then registered onto the functional space where a bounding box is constructed ([Fig. 1](#)) and used to crop the SBRef image and the fMRI volume.

The model uses static-temporal fusion via FiLM² conditioning to learn a probability distribution for a voxel containing CSF inflow signal. Its architecture is shown in [Figure 2](#), with inputs:

1. SBRef: voxel SBRef magnitude, min-max normalized to [0, 1] within the bounding box;
2. Position Coordinates: normalized horizontal distance from center of bounding box (-1 = left-most, +1 = right-most); normalized vertical distance from center of bounding box (-1 = upper-most, +1 = bottom-most); slice number (1 for bottom slice, up to slice 4);
3. Age Group: 0 for ages 18-39, 1 for ages 60-83;
4. Percent Signal Change Timeseries: percent signal change of the voxel's fMRI signal relative to its 15th-percentile baseline, combining the first and last minute of the time series (60 seconds / 0.378 s per TR $\approx$ 158 time points per minute, 316 total)

The final output is a probability map over voxels within the bounding box across the bottom 4 slices of the fMRI volume.

Results

Our model achieved 80.2% F1, 84.1% precision, and 82.4% recall. Example predictions are shown in [Figure 3](#). Performance was evaluated with 10-fold cross-validation, holding out 10% of participants per fold for testing. Within each training fold, 20% of participants were further reserved for early stopping (average precision, patience = 1 epoch). Models were trained with Adam (learning rate = 0.001, batch size = 256) using binary cross-entropy loss, ensuring that every participant is tested only on a model that has never seen their data.

We then validated that model-selected voxels follow known physiological patterns. Prior work has shown that the CSF inflow signals has more power in low frequencies (0.01-0.1 Hz) during NREM sleep compared to the wake state¹, and this difference is more prominent in younger people than in older people.³ We replicated this analysis using our new automated segmentation and successfully reproduced both this aging effect and sleep effect ([Fig. 4](#)). Second, we expect high-intensity visual stimulation (e.g. flickering checkerboard patterns) to drive CSF inflow into the fourth ventricle.⁴ The automated mask also successfully reproduced this result ([Fig. 5](#)).

Discussion

This model automatically selects fMRI voxels in the fourth ventricle containing CSF inflow signal. We hope our system can be readily applied to a wide range of existing datasets, provided that appropriate adaptation is done for datasets where image acquisition parameters differ. Our automated voxel selection can reproduce known physiological properties of CSF inflow across arousal states, age-groups, and in visual stimulation. By removing the need for manual labeling in large datasets, our system lowers the barrier for researchers to measure CSF dynamics.

Acknowledgment

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References

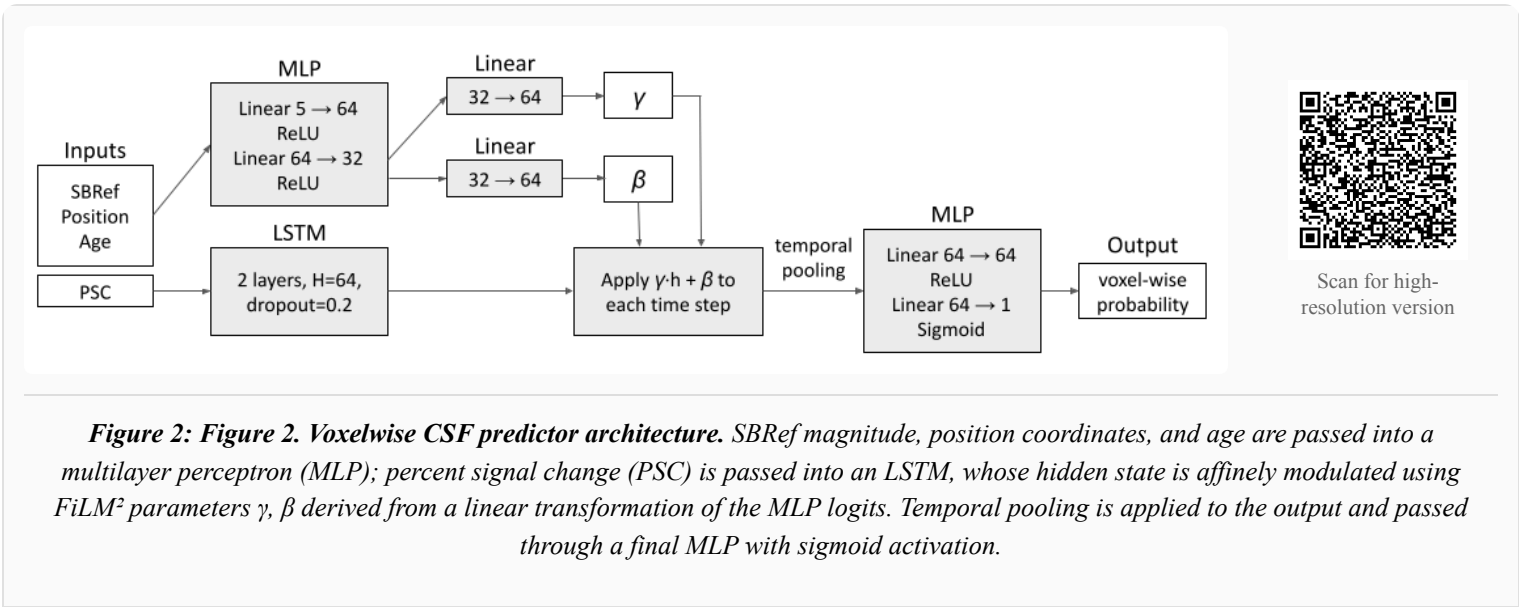
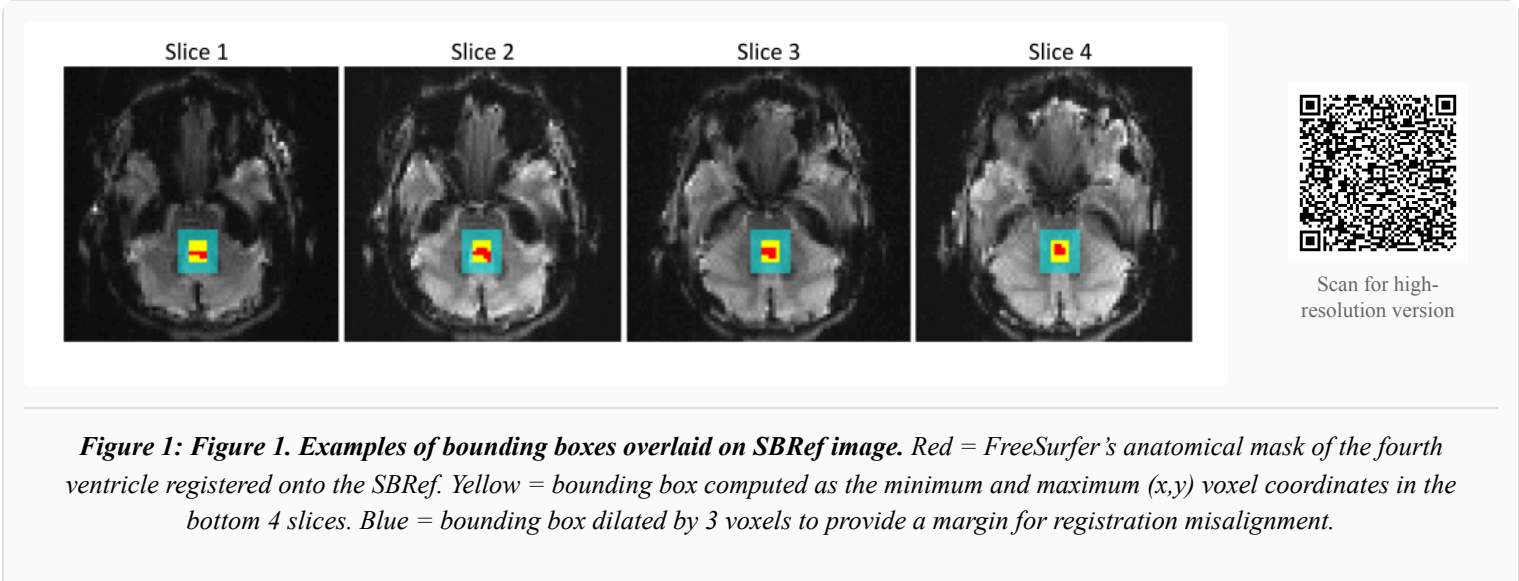
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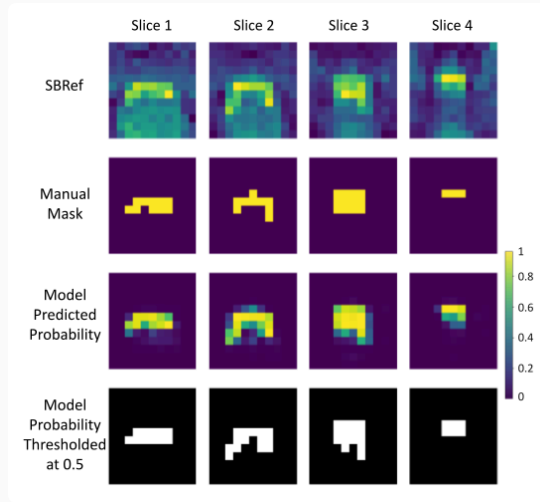
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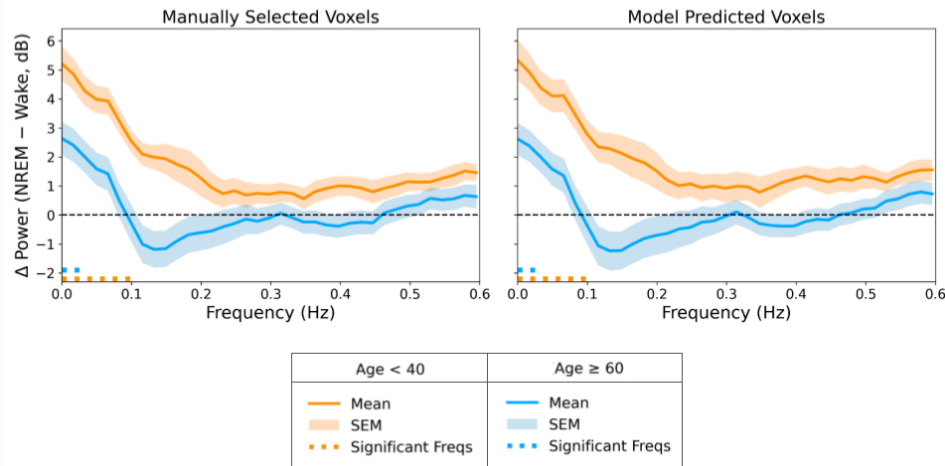
Figures and Tables





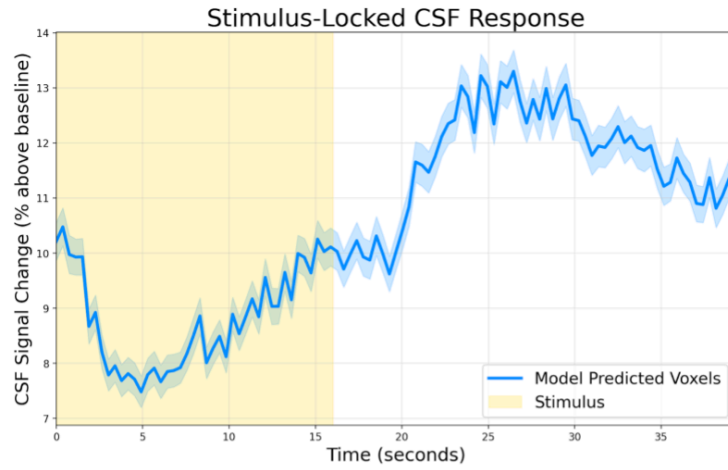
Scan for high-resolution version

Figure 3: Figure 3. Example prediction results. This particular scan achieved 80.1% F1, representative of the average F1 performance across the training data.



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Figure 4: Figure 4. CSF flow power increases during NREM sleep relative to wakefulness. Solid lines show mean NREM–wake spectral difference ($n = 35$, age < 40; $n = 42$, age ≥ 60). Model voxels (probability ≥ 0.5) recover the same significant frequencies in 0–0.1 Hz band with larger NREM than wake power as manual voxels ($p < 0.05$, paired t -tests, BH-FDR). This effect is larger in younger than older participants (group mean difference $\pm$ inter-subject STD = 2.54 ± 4.21 dB for manual; 2.73 ± 4.56 dB for model).



Scan for high-resolution version

Figure 5: Figure 5. Stimulus-locked CSF flow response. Solid line represents mean CSF signal, relative to the 15th percentile baseline, across repetitions of the checkerboard stimulus block experiment (78 subjects, 286 runs, 1633 trials). Shading represents SEM across trials. Checkerboard stimulus is on for the first 16 seconds, then turns off. As expected, CSF flow signal decreases during the stimulus (when blood volume increases) and increases after the stimulus turns off (as blood volume drops).⁴